Auditing SHAP Explanations Against Latent Failure Mechanisms in Predictive Maintenance

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***Abstract*—Machine learning models are increasingly used to predict equipment failures in industrial systems, helping reduce downtime and maintenance costs. However, many high-performing models operate as black boxes, making it difficult to understand their predictions. Explainable AI (XAI) methods identify influential features, but whether these explanations align with the underlying physical failure mechanisms remains unclear. To investigate this gap, we propose a statistical framework that evaluates the consistency between feature attributions and known failure mechanisms using three complementary measures. Using the AI4I 2020 dataset, which provides both failure outcomes and mechanism-specific labels, we assess whether observed alignment exceeds chance-level expectations through comparisons with 1,000 models trained on randomly shuffled labels. Experiments show that an XGBoost classifier (test F1 = 0.89) achieves statistically significant alignment for three failure mechanisms across five random seeds and against a null re-tuned on every permutation, with supporting results from a random forest and interventional TreeSHAP; a neural network explained by Integrated Gradients or KernelSHAP shows weaker alignment. However, explanations for tool wear are sensitive to the choice of SHAP formulation, which we trace mainly to the reference distribution against which attributions are measured. Controlled experiments further show that models can achieve strong predictive performance while assigning substantial importance to misleading features, highlighting a potential disconnect between predictive accuracy and explanation reliability. Additionally, some evaluation measures become less sensitive to attribution differences as dataset size increases. These findings demonstrate that predictive accuracy alone cannot establish the physical relevance of model explanations and underscore the importance of statistically evaluating their alignment with known failure mechanisms.**

***Index Terms*—Explainable AI, SHAP, predictive maintenance, feature attribution evaluation, physical failure mechanisms**

I. Introduction

Predictive maintenance is a maintenance strategy that uses machine-condition data to help anticipate faults and determine when maintenance should be performed. By identifying developing problems, it can help reduce unplanned downtime [1]. Machine-learning models support this process by analyzing equipment data to predict failures and maintenance needs. These models include decision trees, random forests, gradient-boosted trees, support vector methods, and neural networks [2], [3]. Complex ensembles and neural networks can be difficult to interpret, making prediction performance alone insufficient for understanding their behavior [4]. Understanding which sensor readings influence their predictions can help maintenance teams assess an alarm and decide what to investigate [5]. Interpretability approaches used in predictive maintenance include decision trees and local surrogate explanations such as Local Interpretable Model-agnostic Explanations (LIME) [2], [6], as well as SHapley Additive exPlanations (SHAP) for interpreting sensor contributions to model predictions [5], [7]. Recent work has also examined how preprocessing and model complexity affect explanation quality in remaining-useful-life prediction [8].

High predictive performance can coexist with reliance on shortcuts or correlations unrelated to the intended task [9], [10]. In predictive maintenance, this raises the question of whether explanations emphasize the expected features for each failure type. SHAP assigns feature contributions to individual predictions [4], but these contributions do not by themselves establish the physical relevance of those inputs [11]. Evaluating feature attributions is difficult when a suitable ground-truth reference is unavailable [12]. Assessing mechanism alignment therefore requires information about which failure mechanism occurred and which features its rule depends on. Label-permutation methods provide a statistical reference for evaluating feature importance when the association between inputs and outcomes is disrupted [13]. The question is therefore not only which features receive attribution, but whether their attribution is consistent with the recorded mechanism and exceeds that observed under this reference.

In this work, we use the AI4I 2020 predictive maintenance dataset, a synthetic dataset containing machine operating

measurements, an overall failure label, and separate failure-type labels [14]. Its recorded failure types and documented generation rules provide an external reference for evaluating mechanism alignment. Previous work on this dataset has assessed explanation quality using selected examples [15], while subsequent work has used SHAP to interpret feature importance [16]. More recently, Obeten and Jimoh have quantitatively compared SHAP and LIME feature rankings on AI4I 2020 [17]. Agreement between explanation methods, however, does not by itself establish alignment with a failure mechanism. We build on these studies by testing mechanism-level attribution on held-out observations against a reference obtained from models retrained on shuffled labels. We also examine whether the audit detects reliance on a deliberately fabricated feature.

Our approach trains a classifier to predict overall machine failure while withholding the individual failure-type labels from its inputs. We use SHAP because it provides individual feature contributions that can be aggregated by failure type and compared with expected trigger features. TreeSHAP supports efficient attribution computation for the XGBoost and Random Forest models audited in this work [18]. We use the recorded failure-type labels to group these explanations by mechanism. From absolute SHAP values, the audit measures three properties: the attribution share assigned to expected trigger features, how much larger that share is for the relevant mechanism than for other failures, and the largest excess attribution share assigned to a feature outside the expected set. Explanations are computed through five-fold cross-fitting, so each audited observation is explained by a model that was not trained on it. For the primary audit, we construct a reference using 1,000 permutations of the overall failure labels, retraining models and recomputing explanations and statistics for each permutation while retaining the recorded mechanism groups. We apply Holm correction across the four physical mechanisms, separately for each statistic.

To extend example-based explanation assessments with a statistical evaluation of mechanism alignment, we make the following contributions:

- We propose a mechanism-level SHAP audit with held-out explanations and statistical testing, showing that a random-subset baseline can misleadingly suggest alignment.
- We assess robustness across data, model, and explainer settings, including a null re-tuned on every permutation and a neural network explained by KernelSHAP and Integrated Gradients. Heat dissipation, power, and overstrain show alignment for tree ensembles, while tool-wear conclusions depend on the reference distribution of the explainer.
- We test fabricated sensors of varying strength, showing that strong predictive performance can conceal shortcut reliance and that weaker shortcuts can escape detection.
- We evaluate detection up to 300,000 observations and runtime up to one million, showing that shortcut flags can coexist with relevant-feature alignment.
- We benchmark seven classifier families with performance confidence intervals and provide code, model artifacts, and reproduction scripts.

II. Related Work and Background

Predictive maintenance studies use decision trees, random forests, and gradient-boosted trees [2], as well as support vector methods and neural networks [3]. Although small decision trees can offer interpretable rules, complex ensembles and neural networks often require additional methods to explain their predictions [4]. This has motivated the use of explanation methods to help maintenance teams understand model behavior. Cummins et al. [1] review approaches ranging from inherently interpretable models to post-hoc methods such as surrogate models, LIME, and SHAP. These methods serve different purposes: local explanations help examine individual predictions, while global summaries describe recurring patterns across observations. In maintenance applications, this distinction matters because explaining an individual alarm and understanding a model's general reliance on sensor measurements are related but different tasks.

Several studies have investigated these methods in maintenance settings. Steurtewagen and Van den Poel [5] combine sensor and report data with an XGBoost model and use Shapley values to support the diagnosis of machinery breakdowns in the petrochemical industry. Their work connects predictive modeling with diagnostic information that can support maintenance and root-cause investigations. Jalali et al. [3] compare LIME, SHAP, and attribute-wise explanations for models trained on the PHM08-CMAPSS sensor dataset, discussing their advantages and limitations for time-series prediction. These studies illustrate both the practical use of explanations and the importance of selecting an explanation method suited to the maintenance task.

Existing predictive maintenance studies have also evaluated whether explanations agree with known failure mechanisms. Matzka [15] assessed explanation quality against known failure mechanisms using selected examples, including five examples from each of four physical failure types. This provides a direct precedent for comparing explanations with the features expected to matter for a failure. Subsequent work has used SHAP to interpret predictions and feature importance on the dataset [16]. Our study extends this direction by evaluating aggregated attribution patterns on held-out observations and comparing them with a reference obtained from models retrained on shuffled labels.

SHAP assigns each input feature a contribution to an individual prediction [4]. Together with a baseline value, these contributions reconstruct the model output being explained. Their signs indicate the direction of their contribution, while their magnitudes indicate its size on the explained output scale. However, "the" Shapley value of a prediction is not unique: Shapley variants differ in how they remove a feature, for example by conditioning on the remaining features, marginalizing over a background distribution, or substituting a single baseline [19], [20]. Recent surveys organize Shapley

estimators by these removal choices and show that they answer different questions about the model and the data [21], [22], and attribution changes with the treatment of feature dependence when inputs are correlated [23], [24]. For tree ensembles, path-dependent TreeSHAP weights each path by its training cover, whereas interventional TreeSHAP averages over an explicit background sample [18]; faster exact algorithms such as Linear TreeSHAP [25] and Quadrature-TreeSHAP [26] reduce the cost of the path-dependent variant but do not change what it measures. This issue is relevant when engineered features share information with the measurements from which they were constructed, and Section V-D shows that the reference distribution can decide a mechanism's verdict. Agreement with expected trigger features must therefore be interpreted in light of the explanation formulation, rather than treated as proof of physical causation.

Broader explanation-evaluation research provides complementary ways to assess attribution quality. XAI-Bench uses synthetic data for controlled evaluation [27], while OpenXAI evaluates properties including faithfulness, stability, and fairness [28]. Perturbation-based approaches such as ROAR remove highly attributed inputs, retrain the model, and measure changes in predictive performance [29]. Dependence-based global sensitivity indices, such as those built on the Hilbert–Schmidt independence criterion (HSIC) [30], quantify how strongly the output depends on each input without a removal step, and could serve as a model-level corroboration of attribution-based conclusions for correlated features. These studies motivate evaluating explanations beyond visual inspection or feature rankings. A complementary question is whether attributed features agree with external knowledge of the failure mechanism. Evaluating this agreement requires an external reference, such as known failure rules, alongside evidence about the model's attribution patterns.

Randomization offers a reference for evaluating whether an observed pattern exceeds what would arise without the original input–outcome relationship. Label-permutation tests have been used to assess classifier performance [31], and outcome permutations have been used to calibrate feature-importance measures [13]. Adebayo et al. [32] further showed that some visually convincing saliency methods can remain insensitive to changes in model parameters or training labels. These findings motivate explicit controls when interpreting explanations. Research on shortcut learning provides another reason for such controls: models can exploit predictive associations that do not represent the intended task mechanism [9], [10].

Together, these studies motivate evaluating both whether explanations agree with known failure mechanisms and whether they reveal unexpected feature reliance. We address these questions through a mechanism-level audit of held-out SHAP attributions, using recorded failure types and their expected trigger features as an external reference. We assess the attribution statistics against models retrained on shuffled overall failure labels and apply correction for multiple comparisons. A controlled fabricated-sensor experiment examines the audit's ability to detect unexpected reliance and identifies cases it misses. This combines mechanism-based assessment with statistical testing and an explicit evaluation of shortcut-detection capability.

III. Methodology

A. Problem Definition

Predictive maintenance models use sensor measurements and operating conditions to predict equipment failures before they occur. However, a model may correctly predict a failure without relying on the physical conditions that caused it. SHAP (SHapley Additive exPlanations) is an explainability method used to identify which features contribute to a model's predictions. However, assigning high importance to a feature does not necessarily mean that the model's explanation agrees with the actual failure mechanism. This raises a key question: *When a particular failure occurs, does the model assign greater importance to the features involved in that failure's physical rule?*

Let $\mathcal{D} = \{(x_i, y_i, z_i)\}_{i=1}^{N}$ denote a dataset where $x_i \in \mathbb{R}^d$ contains the input features, $y_i \in \{0, 1\}$ indicates overall equipment failure, and $z_i \in \{0, 1\}^M$ records M individual failure mechanisms. A classifier f is trained using only (x_i, y_i); the mechanism indicators are reserved for evaluating its explanations.

For each mechanism m, domain knowledge identifies a set E_m of *trigger features* involved in its physical rule. We define $S_m = \{i : z_{i,m} = 1\}$ as the observations where mechanism m occurs and $F = \{i : y_i = 1\}$ as all observations labeled as failures. Let $\phi_j(x_i)$ denote the SHAP attribution of feature j for observation i.

Our audit evaluates three properties: (i) whether attribution is concentrated on the expected trigger features, (ii) whether that concentration is greater for the corresponding mechanism than for other failures, and (iii) whether unrelated features receive unusually high attribution. These tests measure agreement with known failure rules, not whether SHAP establishes the physical cause of failure. Figure 1 summarizes the data flow of the audit, detailed in the remainder of this section.

B. Dataset and Failure Mechanisms

AI4I 2020 [15] is a synthetic industrial predictive maintenance dataset representing a milling process. It contains 10,000 observations, each recording product quality (`L`, `M`, or `H`), air and process temperatures, rotational speed, torque, tool wear, an overall failure label, and five failure-mechanism indicators. Only 339 observations (3.4%) are labeled as failures, making the prediction task highly imbalanced.

The dataset describes five mechanisms: tool wear failure (TWF), heat dissipation failure (HDF), power failure (PWF), overstrain failure (OSF), and random failure (RNF). The overall failure label is intended to follow the logical OR of these mechanisms:

$$y_i^{\mathrm{OR}} = \mathbb{I}\left(\sum_{m=1}^{M} z_{i,m} \geq 1\right), \tag{1}$$

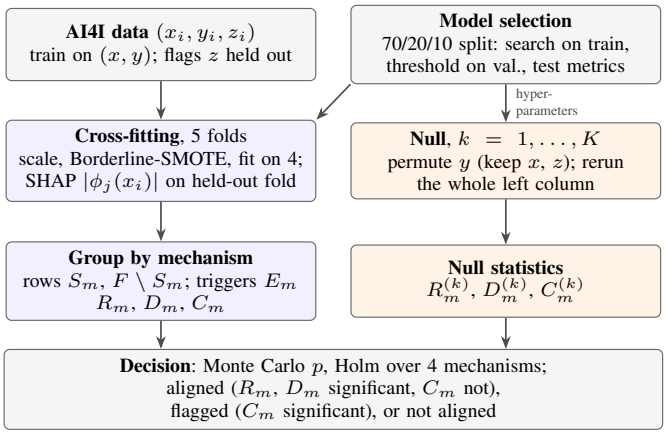


Fig. 1. Audit pipeline. Hyperparameters are selected once on the training split; the audit then explains every observation with a model that did not train on it (blue) and repeats the identical procedure on K label permutations (orange). Mechanism flags z are used only to group the explanations.

TABLE I
AI4I 2020 FAILURE MECHANISMS [15], THEIR TRIGGER CONDITIONS, AND EXPECTED FEATURES E_m. COUNTS ARE FROM THE RELEASED DATASET. THE TWF FAILURE POINT IS RANDOM WITHIN THE SPECIFIED WEAR INTERVAL.

Mech.	Trigger	E_m	n
TWF	tool wear at a random point in 200–240 min	tool wear	46
HDF	(process − air temp.) $<$ 8.6 K and speed $<$ 1380 rpm	both temps., their difference, speed	115
PWF	torque × speed (rad/s) outside [3500, 9000] W	torque, speed, power	95
OSF	tool wear × torque above 11,000/12,000/ 13,000 min·Nm for L/M/H	tool wear, torque, overstrain, its ratio	98
RNF	0.1% chance, independent of the process	–	19

Note: E_m denotes the expected trigger-feature set for mechanism m, and n is the number of observations with that mechanism. L, M, and H denote product quality types. RNF has no defined physical trigger features.

where $\mathbb{I}(\cdot)$ equals one when at least one mechanism is active. Thus, the overall label indicates whether failure occurred, while the mechanism indicators describe which failure conditions were recorded. Table I lists the documented rules and the corresponding trigger features used in our audit.

a) Overlapping mechanisms and label discrepancies: The released annotations do not always follow the intended OR relationship. Twenty-four observations have multiple active mechanism flags; these are valid overlapping conditions rather than labeling errors. However, nine observations are labeled as failures without any active flag, while 18 of the 19 RNF-flagged observations have an overall failure label of zero.

We retain the released overall labels for training rather than reconstructing them from the mechanism flags. The primary audit includes every observation where the mechanism being evaluated is active, including overlapping cases. We also repeat the analysis using observations with exactly one active mechanism flag, removing ambiguity about which mechanism may explain the attribution. Comparing both analyses shows whether overlapping conditions affect the conclusions. RNF has no physical trigger features and is examined only descriptively.

C. Feature Construction

Several mechanisms depend on combinations of sensor readings rather than individual measurements. We therefore construct four features from the original inputs: mechanical power $P = \tau(2\pi n/60)$, temperature difference $\Delta T = T_{\text{process}} - T_{\text{air}}$, overstrain $O = w\tau$, and overstrain ratio $Q = O/L(t)$, where τ, n, and w denote torque, rotational speed, and tool wear, respectively. The type-specific limit $L(t)$ is 11,000, 12,000, or 13,000 for product types `L`, `M`, and `H`.

These features explicitly represent the interactions used in the HDF, PWF, and OSF rules. They are calculated from process measurements and product type, without using failure labels or mechanism flags. However, they are correlated with the original measurements and encode known parts of the failure rules. We therefore repeat the audit using only the original features to determine how much the results depend on feature construction (Section V-A).

D. Explanation Audit Statistics

We evaluate attribution using three complementary statistics. For feature j and observation set A, let

$$I_j(A) = \frac{1}{|A|} \sum_{i \in A} |\phi_j(x_i)|, \qquad s_j(A) = \frac{I_j(A)}{\sum_{k=1}^{d} I_k(A)} \tag{2}$$

denote the average absolute SHAP contribution and the attribution share of feature j, where $\phi_j(x_i)$ is computed by the cross-fitted model that did not train on observation i (Section III-E). We use absolute values because positive and negative contributions could otherwise cancel. The fraction of attribution assigned to the expected trigger features of mechanism m is

$$r_m(A) = \sum_{j \in E_m} s_j(A). \tag{3}$$

From these quantities, we define three statistics:

$$R_m = r_m(S_m), \tag{4}$$

$$D_m = r_m(S_m) - r_m(F \setminus S_m), \tag{5}$$

$$C_m = \max_{j \notin E_m} \left[s_j(S_m) - s_j(F \setminus S_m) \right], \tag{6}$$

All three are computed on the same matrix of held-out absolute SHAP values; only the observation sets and feature sets differ.

The *relevant score* R_m measures how much attribution goes to the expected trigger features on mechanism m's observations. The *specificity score* D_m asks whether those features receive a larger share for this mechanism than for other labeled failures. The *shortcut score* C_m measures the

TABLE II
THE THREE AUDIT STATISTICS AND THEIR INTERPRETATION.

Statistic	Question	Desired result
R_m, relevant share	How much attribution goes to the expected trigger features?	High
D_m, specificity	Is that share higher for this mechanism than for other failures?	High
C_m, shortcut	Does a non-trigger feature gain unusually high attribution?	Not high

largest mechanism-specific increase in attribution among features outside the expected trigger set. A large C_m raises concern about reliance on another feature, although it does not prove that the feature is causally irrelevant.

For example, among the 95 PWF observations, the average absolute SHAP contributions of power, torque, and rotational speed are 4.96, 1.93, and 1.33, respectively (Fig. 3a). Their sum is 8.22, compared with 11.82 across all twelve inputs, giving $R_{\mathrm{PWF}} = 8.22/11.82 \approx 0.70$. Thus, approximately 70% of the attribution on PWF observations goes to features involved in its physical rule. Whether this agreement exceeds what could occur by chance is determined through statistical testing.

E. Out-of-Sample Explanations

Explanations calculated on training observations may reflect patterns the model has fitted particularly closely to those observations. We therefore use five-fold cross-fitting [33]. The data are divided into five stratified folds. In each round, the complete training pipeline is fitted on four folds, and SHAP values are calculated on the held-out fold. After five rounds, every observation has an explanation from a model that was not trained on it. We combine these held-out explanations for the audit and also examine the effect of using in-sample SHAP values (Section V-A).

F. Statistical Significance

A high attribution score does not necessarily indicate meaningful alignment. Mechanism-specific observations often contain extreme sensor values, which can receive large SHAP contributions even when the model has learned no genuine relationship with failure. This makes a simple comparison against randomly selected failure observations unreliable (Section V-B).

We instead construct a *shuffled-label null*. For each null model, we randomly permute the overall failure labels while keeping the sensor measurements and mechanism indicators unchanged. This preserves the input distribution and failure rate but removes the original relationship between inputs and failure outcomes. We repeat the full preprocessing, oversampling, training, cross-fitting, and explanation procedure, using the hyperparameters selected on the original data. Because carrying these hyperparameters over could itself shape the null, we also rebuild it with the full hyperparameter search repeated on every permutation (Section V-A).

Repeating this process K times provides a reference distribution for each statistic. For an observed value T and null values $T^{(1)}, \ldots, T^{(K)}$, we calculate the upper-tail Monte Carlo p-value:

$$p = \frac{1 + \sum_{k=1}^{K} \mathbb{I}(T^{(k)} \geq T)}{K + 1}. \tag{7}$$

The primary audit uses $K = 1{,}000$, giving a minimum possible p-value of $1/1001 \approx 0.001$. We apply Holm's correction [34] across the four physical mechanisms for each statistic and use $\alpha = 0.05$. To complement the p-values, we report a null-standardized effect size $z = (T - \bar{T}_0)/s_0$, where $\bar{T}_0$ and s_0 are the mean and standard deviation of the null values, and 95% percentile bootstrap intervals for R_m and D_m obtained by resampling observations within S_m and $F \setminus S_m$ (5,000 resamples). The bootstrap holds the cross-fitted models fixed, so it reflects observation sampling only; variability from refitting is captured by the null.

We combine the three tests into one decision rule. A mechanism is *aligned* when both R_m and D_m are significantly larger than their null distributions and C_m is not; it is *flagged* when C_m is significant, whatever R_m and D_m show, and *not aligned* otherwise. The shortcut test can therefore overturn a positive verdict, which matters at large n, where a narrow null lets R_m pass even when a non-trigger feature takes much of the attribution (Section V-E). A flag names the responsible feature and its excess share, which should be inspected before acting on it. A non-significant shortcut result means that the audit did not detect such an increase; it does not establish that shortcuts are absent. RNF is excluded from the physical-alignment tests because it has no expected trigger features.

G. Model Training and Evaluation

We split the dataset into stratified training, validation, and test sets (70/20/10). Numerical scaling and one-hot encoding are fitted on training data only. Because failures account for just 3.4% of observations, we compare SMOTE [35], Borderline-SMOTE [36], and ADASYN [37] using the validation F1 of a default XGBoost model [38]. Borderline-SMOTE gives the highest validation F1 and is used thereafter.

We compare seven classifiers: logistic regression, an RBF-kernel SVM, k-NN ($k = 7$), a decision tree (depth 10), a random forest (300 trees, depth 12), XGBoost (300 trees, depth 6, learning rate 0.1), and an MLP with hidden layers of 64 and 32 units. Other hyperparameters use library defaults (scikit-learn 1.9.1, XGBoost 3.4.1, and PyTorch 2.14.0), with random seed 42.

XGBoost achieves the highest validation F1. We tune it using 40 iterations of randomized search with five-fold stratified cross-validation, optimizing F1. The search covers tree depth 3–8, 200–600 trees, learning rate 0.01–0.2, row and column subsampling 0.6–1.0, minimum child weight 1–5, and γ 0–0.3. Borderline-SMOTE is applied within each training fold to avoid oversampling leakage. We select the failure-probability threshold that maximizes validation F1.

TABLE III
EXPERIMENTAL PLAN. K IS THE NUMBER OF SHUFFLED-LABEL MODELS USED FOR EACH STATISTICAL COMPARISON.

Experiment	Question	K
Primary audit, seed 42	Do explanations match known mechanisms? (H1)	1,000
Re-tuned null	Does carrying over hyperparameters bias the null?	200
Seeds 1–4	Are results stable across random seeds?	200
Single-flag rows	Do overlapping mechanisms affect the results?	1,000
Raw features only	Do engineered features influence alignment?	1,000
In-sample SHAP	Does explaining training observations change results?	1,000
Interventional TreeSHAP	Does the explanation method or its background affect the audit?	200
Random forest	Does the model family affect alignment?	200
MLP with KernelSHAP / IG	Does the audit extend beyond trees and TreeSHAP?	200
Shuffled-label model	Does the naive null give misleading results? (H3)	200
Shortcut sweep	When is a misleading feature detected? (H2)	500
Scaling, 10^4–$3 \cdot 10^5$ rows	How do sensitivity and cost change with size?	200

We evaluate predictive performance using failure-class precision, recall, F1, ROC-AUC, Brier score, and expected calibration error. We report 95% confidence intervals using exact binomial (Clopper–Pearson) intervals for precision and recall and 5,000 stratified bootstrap resamples for other metrics. We additionally report five-fold cross-validation results at threshold 0.5. Because this evaluation reuses data involved in hyperparameter selection, its estimates may be slightly optimistic.

H. Experimental Design

We conduct the experiments in Table III to examine whether the audit detects physical alignment, how stable its results are, and whether it identifies misleading attribution. The primary audit uses 1,000 shuffled-label models and takes approximately 12 minutes on eight CPU cores.

a) Robustness checks: We repeat the audit across four additional random seeds, on single-mechanism observations, without engineered features, and using in-sample rather than held-out SHAP values. We also compare path-dependent TreeSHAP with interventional TreeSHAP, using background samples of 100, 500, and 1,000 training observations per fold, drawn uniformly at random from that fold's original (not oversampled) training observations with up to three background seeds, and repeat the audit with a random forest. To examine the TWF result, we further draw the interventional background only from non-failures, only from failures, or from the oversampled training data, and run the interventional audit on the original features alone. Finally, to test whether the audit depends on tree ensembles or TreeSHAP, we audit the MLP of Section III-G with two non-tree explainers: KernelSHAP [4], a model-agnostic Shapley estimator (25 training observations as background, 1,000 coalitions per observation), and Integrated Gradients [39], a gradient-based method (baseline equal to the mean of 100 training observations, 1,024 steps, completeness checked to within 5%). Both explain the MLP's logit, and every null model retrains the MLP on permuted labels and is explained in the same way. Each alternative uses a matching shuffled-label null procedure.

b) Controlled shortcut experiment: To test whether the audit detects reliance on an unrelated but predictive feature, we introduce an artificial sensor

$$s_i = z_{i,\mathrm{HDF}} + \epsilon_i, \qquad \epsilon_i \sim \mathcal{N}(0, \sigma^2). \tag{8}$$

This sensor is a noisy copy of the HDF indicator, created only for this controlled experiment. It has no physical role in heat dissipation failure and is excluded from E_{HDF}. We decrease σ from 2 to 0.1; smaller values make the artificial sensor more predictive. We examine whether the audit detects increasing reliance on this shortcut.

c) Scaling experiment: We generate new observations using the documented AI4I 2020 process [15], without copying released rows. The sensor distributions follow the documented process, and HDF, PWF, and OSF use their published rules. Tool wear increases until a lifetime drawn from $U(200, 240)$ min, where TWF occurs with probability 0.4. The generated failure rate is 3.5–4.0%, compared with 3.4% in the release.

We evaluate $n \in \{10^4, 3 \cdot 10^4, 10^5, 3 \cdot 10^5\}$ with three generation seeds for the first three sizes and one for the largest. Each setting is tested without a shortcut and with shortcut noise $\sigma \in \{0.35, 0.25\}$. The scaling experiments use $K = 200$ null models, and the generator is included in the released code.

d) Why simulated data: We use AI4I 2020 because its documented failure mechanisms provide a reference for evaluating explanation alignment, which is rarely available in real maintenance datasets. Although the data are generated from known rules, the classifier is trained only on overall failure labels and may learn alternative predictive relationships. Agreement with the physical rules is therefore not automatic.

The shuffled-label, shortcut, and raw-feature experiments test whether the audit can distinguish mechanism-consistent attribution from chance agreement or reliance on alternative features. Nevertheless, the findings establish agreement with known simulated failure rules, not causal validity in real industrial systems.

IV. EXPERIMENTAL RESULTS

A. Predictive Performance

We first evaluate the classifiers on the binary failure-prediction task. Borderline-SMOTE achieves the highest validation F1 (0.826), compared with 0.768 for SMOTE and 0.747 for ADASYN, and is therefore used in subsequent experiments.

Table IV compares the seven classifiers. The tuned XGBoost model achieves the highest test F1 of 0.89, compared with

TABLE IV
FAILURE-CLASS TEST PERFORMANCE, SORTED BY F1. PARENTHESIZED INTERVALS INDICATE 95% BOOTSTRAP CONFIDENCE INTERVALS FOR F1.

Model	Prec.	Rec.	F1 (95% CI)	ROC-AUC
XGBoost (tuned)	**1.00**	0.79	**0.89** (0.79–0.95)	0.988
XGBoost	0.86	0.88	0.87 (0.78–0.94)	0.988
Random forest	0.79	0.88	0.83 (0.74–0.92)	**0.989**
MLP	0.65	**0.91**	0.76 (0.67–0.84)	0.988
Decision tree	0.68	0.82	0.75 (0.64–0.84)	0.927
SVM	0.42	0.88	0.57 (0.48–0.65)	0.976
k-NN	0.43	0.79	0.56 (0.46–0.65)	0.877
Logistic regression	0.24	0.85	0.37 (0.31–0.43)	0.943

0.87 for the default XGBoost. Its validation-selected decision threshold is 0.978. However, the confidence intervals overlap, so this difference does not establish a reliable improvement from tuning.

Of the 34 failures in the test set, the tuned model correctly identifies 27 and misses seven, without producing a false alarm. This gives precision 1.00 (95% CI: 0.87–1.00) and recall 0.79 (95% CI: 0.62–0.91). Its Brier score is 0.0090, compared with 0.033 for a constant base-rate predictor, and its expected calibration error is 0.008.

Five-fold cross-validation produces a failure-class F1 of 0.822 ± 0.013 and macro-F1 of 0.908 ± 0.007. Although these results indicate that the model can identify failures reasonably well, predictive performance alone does not show whether its explanations agree with the corresponding failure mechanisms.

B. Alignment with Failure Mechanisms

We evaluate the tuned XGBoost model using out-of-sample SHAP values and 1,000 shuffled-label models. The audit examines whether attribution is concentrated on the expected trigger features (R_m), whether this concentration is specific to the corresponding mechanism (D_m), and whether unrelated features receive unusually high attribution (C_m). Figure 2 compares the observed relevant and specificity scores with their shuffled-label distributions.

a) HDF, PWF, and OSF: All three mechanisms pass the relevant and specificity tests, with Holm-adjusted $p = 0.004$ for each. Their trigger features receive substantially more attribution on the corresponding mechanism observations than on other failures, with specificity scores between 0.33 and 0.43 and effect sizes of $z = 5.7$–8.4. The relevant-share effect is smallest for HDF ($z = 3.4$): its four trigger features include both temperatures and rotational speed, on which even shuffled-label models place much attribution (null mean 0.56). The bootstrap intervals are narrow (half-widths of 0.02–0.03) compared with the null standard deviations (0.04–0.06), so uncertainty in these verdicts comes mainly from model refitting rather than from the finite number of mechanism observations. No statistically significant shortcut is detected in the primary audit.

b) TWF: Tool wear failure does not pass either alignment test ($p_R = 0.201$, $p_D = 0.077$). Although tool wear is its most important feature, it receives only 19% of the attribution on TWF observations, compared with 15% under the null. The documented TWF mechanism includes a random failure point within a tool-wear interval, and only 46 observations carry this flag. Both factors may make its attribution harder to distinguish from the null.

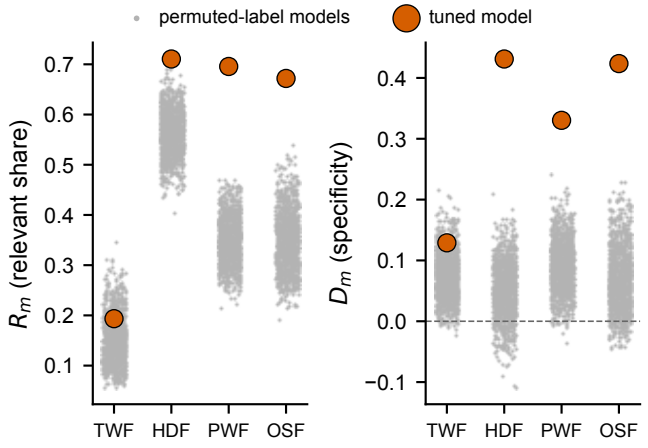


Fig. 2. Primary audit of the tuned XGBoost model. The observed relevant and specificity scores are compared with 1,000 shuffled-label models. HDF, PWF, and OSF exceed the null distributions on both measures, whereas TWF does not.

C. Feature Attribution Across Mechanisms

Figure 3 compares SHAP attribution by failure mechanism with the conventional global summary. The mechanism-level heatmap shows higher attribution to rotational speed and temperature difference for HDF, power and torque for PWF, and overstrain-related features for OSF. In contrast, the global SHAP summary ranks features across all observations without distinguishing the failure mechanisms.

These results show why overall feature importance is insufficient for evaluating mechanism alignment. A feature can be important to the classifier without explaining the particular failure mechanism being examined.

V. EXPERIMENTAL ANALYSIS

We next examine how the audit responds to changes in the data, model, explanation method, and statistical comparison. The results are summarized in Table VI.

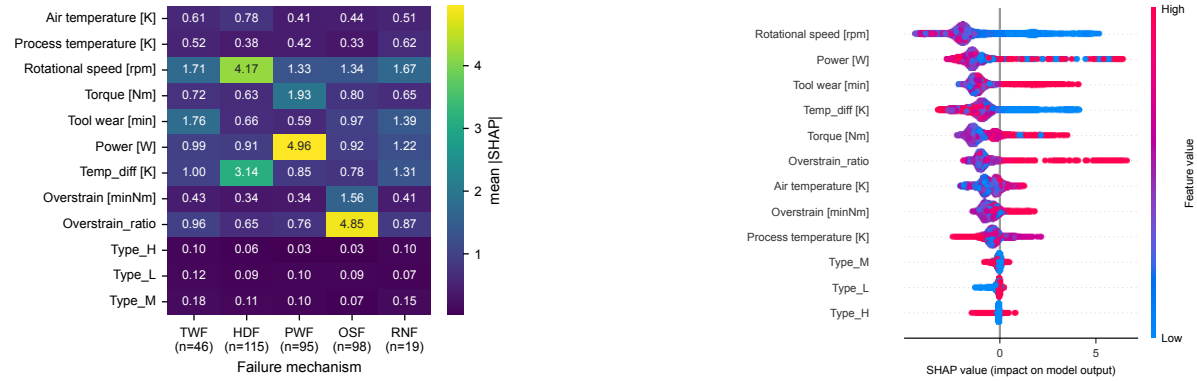
A. Robustness of Mechanism Alignment

a) Random seeds and overlapping mechanisms: HDF, PWF, and OSF pass both alignment tests across all five random seeds and when the analysis is restricted to observations with a single active mechanism. TWF fails both tests across the five seeds. This suggests that the main findings are not driven by a particular random seed or by overlapping mechanism indicators.

TABLE V
PRIMARY AUDIT USING CROSS-FITTED SHAP VALUES (SEED 42, $K = 1{,}000$).

	n	Top feature	R_m [95% CI]	Null	z	p	D_m [95% CI]	Null	z	p	C_m	Null	p
TWF	46	tool wear	0.193 [0.17, 0.22]	0.152	0.9	0.201×	0.129 [0.10, 0.16]	0.070	1.5	0.077×	0.025	0.025	1.000
HDF	115	speed	0.711 [0.69, 0.73]	0.557	3.4	0.004	0.431 [0.41, 0.45]	0.049	8.4	0.004	0.001	0.015	1.000
PWF	95	power	0.696 [0.67, 0.72]	0.347	7.6	0.004	0.330 [0.30, 0.36]	0.088	5.7	0.004	0.001	0.011	1.000
OSF	98	overstrain ratio	0.672 [0.66, 0.69]	0.345	5.6	0.004	0.424 [0.40, 0.44]	0.067	7.0	0.004	-0.001	0.019	1.000
RNF	19	speed	0.186	0.240	–	0.985	–	–	–	–	–	–	–

Note: R_m measures the attribution share assigned to the expected trigger features of mechanism m; D_m measures the difference in that share between mechanism-m failures and other failures; C_m measures the largest mechanism-specific attribution increase among non-trigger features. K denotes the number of shuffled-label models. Null values are averages across these models; z is the observed value minus the null mean, divided by the null standard deviation. Brackets give 95% bootstrap intervals over observations with the models held fixed. Reported p-values are Holm-adjusted across the four physical mechanisms. Significant R_m and D_m ($p < 0.05$) indicate statistical alignment; significant C_m indicates potential shortcut reliance.



(a) Mean absolute SHAP by failure mechanism

(b) Global SHAP summary

Fig. 3. SHAP explanations of the tuned classifier. (a) Attribution patterns for individual failure mechanisms. (b) Global feature importance across all observations. The mechanism-level view reveals distinctions that are not visible in the global ranking.

b) Re-tuned null: Repeating the 40-iteration hyperparameter search on each of 200 permutations gives a wider null than carrying over the hyperparameters selected on the real labels: the search on shuffled labels mostly selects shallow trees (depth at most 4 in 115 of 200 permutations), and the null 95th percentile of R_{OSF} rises from 0.46 to 0.62. The verdicts are unchanged. HDF, PWF, and OSF pass both tests and TWF fails both ($p_R = 0.453$, $p_D = 0.368$), but the OSF relevant-share margin narrows: 4 of 200 re-tuned null models reach the observed 0.672, giving an adjusted $p = 0.0498$. The carried-over null is therefore the less conservative reference, and the OSF relevant-share result should be read as borderline under the stricter one.

c) In-sample versus out-of-sample explanations: When explanations are calculated on observations used for training, all four mechanisms pass, including TWF ($p = 0.004$). Its relevant score increases from 0.19 with out-of-sample SHAP to 0.35 with in-sample SHAP. This difference shows that evaluating training observations can produce a more favorable alignment assessment. We therefore use cross-fitting for the primary audit.

d) Original features only: Removing the engineered features changes the verdict for HDF, which retains significant specificity but no longer passes the relevant-share test. PWF and OSF continue to pass both tests, although OSF specificity is close to the threshold ($p = 0.048$). TWF still fails. Thus, alignment for PWF and OSF is not dependent on explicitly providing the derived features, whereas the HDF result is sensitive to feature construction. This is consistent with the difficulty of representing a temperature-difference rule using only separate temperature measurements in a tree model.

B. Why the Choice of Null Matters

A statistical comparison should reject a model that has learned no genuine relationship between sensor measurements and failures. Our shuffled-label null satisfies this check: a classifier trained on shuffled failure labels fails the alignment tests. However, when we compare that same classifier against randomly selected failure observations under the same model, all four mechanisms appear significant (adjusted $p = 0.002$). The naive comparison therefore approves a model trained on meaningless labels. The likely explanation is that mechanism observations contain unusual sensor values that can receive high SHAP attribution even without meaningful failure learning. This result supports using retrained shuffled-label models rather than relying on differences between observation groups alone.

C. Can the Audit Detect a Shortcut?

We introduce a synthetic feature that is correlated with the HDF indicator but has no physical role in heat dissipation failure. By reducing its noise, we make this feature increasingly

TABLE VI
HOLM-ADJUSTED AUDIT p-VALUES ACROSS EXPERIMENTAL VARIANTS AND FAILURE MECHANISMS.

		TWF			HDF			PWF			OSF		
Variant	K	R	D	C	R	D	C	R	D	C	R	D	C
Primary, seed 42	1000	$0.201^{\times}$	$0.077^{\times}$	1.000	0.004	0.004	1.000	0.004	0.004	1.000	0.004	0.004	1.000
single-mechanism rows	1000	$0.187^{\times}$	$0.076^{\times}$	1.000	0.004	0.004	1.000	0.004	0.004	1.000	0.004	0.004	1.000
null re-tuned per permutation	200	$0.453^{\times}$	$0.368^{\times}$	1.000	0.020	0.020	1.000	0.020	0.020	1.000	0.0498	0.020	1.000
Seed 1	200	$0.199^{\times}$	$0.119^{\times}$	0.776	0.020	0.020	1.000	0.020	0.020	1.000	0.020	0.020	1.000
Seed 2	200	$0.149^{\times}$	$0.065^{\times}$	0.498	0.020	0.020	1.000	0.020	0.020	1.000	0.020	0.020	1.000
Seed 3	200	$0.174^{\times}$	$0.095^{\times}$	0.736	0.020	0.020	1.000	0.020	0.020	1.000	0.020	0.020	1.000
Seed 4	200	$0.269^{\times}$	$0.129^{\times}$	1.000	0.020	0.020	1.000	0.020	0.020	1.000	0.020	0.020	1.000
Raw features only	1000	$0.345^{\times}$	$0.616^{\times}$	1.000	$0.190^{\times}$	0.004	1.000	0.004	0.004	1.000	0.004	0.048	1.000
In-sample SHAP	1000	0.004	0.004	1.000	0.004	0.004	1.000	0.004	0.004	1.000	0.004	0.004	1.000
Interventional, background 100	200	0.020	0.020	1.000	0.020	0.020	1.000	0.020	0.020	1.000	0.020	0.020	1.000
Interventional, background 500	200	0.020	0.020	1.000	0.020	0.020	1.000	0.020	0.020	1.000	0.020	0.020	1.000
Interventional, background 1000	200	0.020	0.020	1.000	0.020	0.020	1.000	0.020	0.020	1.000	0.020	0.020	1.000
failure-only background	200	$0.995^{\times}$	$0.945^{\times}$	$0.020^{\dagger}$	$0.826^{\times}$	$0.149^{\times}$	1.000	0.045	$0.090^{\times}$	1.000	0.040	0.020	1.000
non-failure background	200	0.020	0.020	1.000	0.020	0.020	1.000	0.020	0.020	1.000	0.020	0.020	1.000
oversampled background	200	0.045	0.020	0.896	0.020	0.020	1.000	0.020	0.020	1.000	0.020	0.020	1.000
raw features only	200	0.020	0.020	1.000	0.020	0.020	1.000	0.020	0.020	1.000	0.020	0.020	0.975
Random forest	200	$0.617^{\times}$	$0.398^{\times}$	1.000	0.020	0.020	1.000	0.020	0.020	1.000	0.040	0.020	1.000
MLP, KernelSHAP	200	$0.975^{\times}$	$0.831^{\times}$	0.975	$0.388^{\times}$	$0.189^{\times}$	1.000	$0.358^{\times}$	0.020	1.000	$0.358^{\times}$	$0.075^{\times}$	1.000
MLP, Integrated Gradients	200	$0.338^{\times}$	$0.100^{\times}$	0.522	0.020	0.020	1.000	0.0498	$0.100^{\times}$	0.478	0.030	0.020	1.000
Permuted labels (sanity)	200	$1.000^{\times}$	$0.731^{\times}$	1.000	$0.199^{\times}$	$0.557^{\times}$	1.000	$1.000^{\times}$	$0.731^{\times}$	1.000	$1.000^{\times}$	$0.731^{\times}$	0.637
Naive null, tuned model	–	0.002	–	–	0.002	–	–	0.002	–	–	0.002	–	–
Naive null, permuted labels	–	0.002	–	–	0.002	–	–	0.002	–	–	0.002	–	–

Note: K denotes the number of shuffled-label models. R, D, and C represent the relevant-share, specificity, and shortcut tests, respectively. All reported values are Holm-adjusted p-values. Significant R and D ($p < 0.05$) indicate alignment; significant C indicates potential shortcut reliance. The superscript $\times$ marks non-significant alignment tests, † marks a significant shortcut test, and "–" indicates not evaluated. The minimum attainable adjusted p-value is $4/(K+1)$. Interventional results use background seed 0; other background seeds yield the same verdicts. Indented interventional rows use a background of 100 observations drawn from the stated training subset.

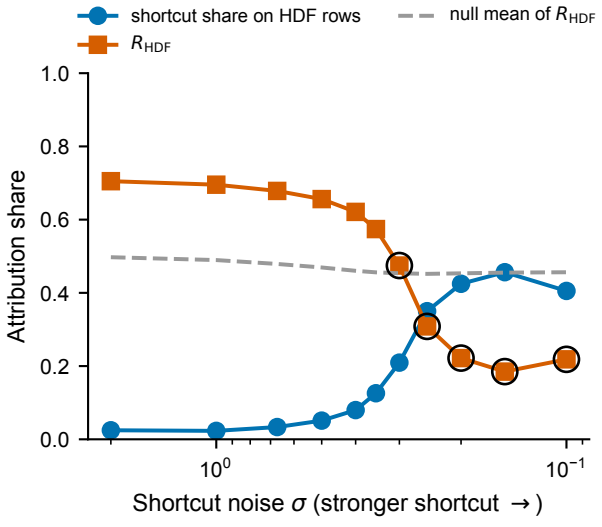


Fig. 4. Effect of an injected shortcut on HDF attribution. As noise decreases, the artificial sensor receives a larger attribution share while the expected trigger features receive less. Markers indicate settings where the relevant-share alignment test fails.

useful for predicting HDF. Figure 4 shows how attribution shifts from the documented trigger features to the artificial sensor.

The audit responds differently depending on the strength of the shortcut.

- **Weak shortcut ($\sigma \geq 0.4$):** The artificial sensor receives at most 8% of attribution, and HDF continues to pass.
- **Moderate shortcut ($\sigma = 0.35$–0.30):** At 13% shortcut attribution, the relevant-share test still passes ($p = 0.044$). When the share reaches 21%, this test fails.
- **Strong shortcut ($\sigma \leq 0.25$):** At approximately 35% shortcut attribution, the shortcut statistic also becomes significant. The specificity test fails once the artificial sensor receives approximately 42%.

Throughout the experiment, test F1 remains between 0.84 and 0.92. Thus, the model can maintain good predictive performance while increasingly relying on a feature unrelated to the documented mechanism.

PWF and OSF continue to pass across the shortcut settings, while the injected feature affects HDF. However, the audit does not reliably detect weak shortcuts, and the shortcut statistic is less sensitive than the relevant-share test at moderate strengths.

D. Effect of the Model and SHAP Method

HDF, PWF, and OSF pass the alignment tests using both path-dependent and interventional TreeSHAP, and their verdicts remain unchanged with a random forest.

a) Beyond trees: The audit itself needs only an attribution matrix, so it runs unchanged on the MLP (test F1 0.76) with KernelSHAP and Integrated Gradients, each against 200 retrained MLP nulls (Table VI). With Integrated Gradients,

HDF and OSF are aligned (adjusted $p \leq 0.030$ for both tests), PWF passes the relevant-share test narrowly ($p = 0.0498$) but not specificity ($p = 0.10$), and TWF fails. With KernelSHAP, no mechanism is aligned; only PWF specificity is significant. Two factors make these verdicts weaker than for XGBoost. First, shuffled-label MLPs already place much attribution on the trigger features, so the null means are high (e.g., R_{PWF} null mean 0.60–0.63 versus 0.35 for XGBoost), likely because a smooth network trained on noise still spreads attribution over the inputs that the PWF and OSF rules use. Second, the MLP is a weaker classifier, and KernelSHAP adds sampling error from its 25-row background and 1,000 of the 4,094 coalitions. We cannot separate these factors here, but the result shows why the shuffled-label null is needed: under KernelSHAP, the raw relevant shares of HDF, PWF, and OSF (0.47–0.67) look like alignment, yet none exceeds what an MLP without failure knowledge produces.

TWF behaves differently. It passes both tests with interventional TreeSHAP across the tested background sizes and seeds, with relevant scores between 0.45 and 0.47. However, it fails with path-dependent TreeSHAP and the random forest. Tool wear is strongly correlated with the engineered overstrain features overall (Spearman 0.92), and different SHAP formulations distribute attribution among dependent features differently [23]. Correlation alone does not explain the flip, however: on TWF observations the correlation is only 0.13, and path-dependent TWF alignment also fails without the engineered features (Table VI). Interventional TreeSHAP also passes TWF on the raw features alone ($R = 0.58$, $p = 0.020$).

The flip is better explained by the reference distribution against which each formulation measures a feature's contribution. Path-dependent TreeSHAP weights tree paths by the training cover, which reflects the oversampled data the model was fitted on, whereas our interventional background is drawn from the original training observations, about 97% of which are non-failures. We therefore repeat the interventional audit with three backgrounds (Table VI), each with a matching null. A non-failure background reproduces the default result ($R_{\mathrm{TWF}} = 0.46$). A background drawn from the oversampled training data lowers R_{TWF} to 0.23, close to the path-dependent 0.19, although TWF still passes narrowly ($p_R = 0.045$). A failure-only background lowers it to 0.08 and also breaks HDF alignment. The attribution therefore answers different questions under different references: against typical operation, tool wear is what distinguishes TWF observations, but against other failures it is not. Thus, the TWF verdict depends mainly on the choice of reference distribution rather than on feature correlation, and remains unresolved in the sense that no single choice is correct for all purposes. For maintenance audits that ask why an observation is abnormal, we recommend a background drawn from normal operation and reporting it alongside the result.

E. Does the Audit Work with More Data?

We repeat the audit on generated datasets containing 10^4 to $3 \cdot 10^5$ observations. Without an injected shortcut, HDF, PWF, and OSF pass the alignment tests at every evaluated size and seed. TWF continues to fail the relevant-share test, showing that additional observations alone do not resolve its alignment.

TABLE VII
AUDIT RESULTS ON GENERATED DATA WITH INCREASING DATASET SIZE ($K = 200$). THE TABLE REPORTS HDF SHORTCUT ATTRIBUTION, DETECTION FREQUENCIES, TWF OUTCOMES, AND RUNTIME. HDF, PWF, AND OSF PASS THE ALIGNMENT TESTS IN EVERY RUN WITHOUT THE INJECTED SENSOR.

				HDF flagged by		TWF		
n	σ	Seeds	Sensor	R	C	D pass	C flag	Min.
10,000	–	3	0.000	0/3	0/3	0/3	0/3	3
	0.35	3	0.158	2/3	0/3	0/3	0/3	3
	0.25	3	0.390	3/3	3/3	0/3	0/3	3
30,000	–	3	0.000	0/3	0/3	0/3	1/3	7
	0.35	3	0.128	0/3	0/3	2/3	0/3	6
	0.25	3	0.346	3/3	3/3	2/3	1/3	7
100,000	–	3	0.000	0/3	0/3	1/3	2/3	17
	0.35	3	0.100	0/3	0/3	2/3	1/3	19
	0.25	3	0.315	3/3	3/3	2/3	1/3	19
300,000	–	1	0.000	0/1	0/1	0/1	0/1	52
	0.35	1	0.063	0/1	0/1	0/1	0/1	59
	0.25	1	0.271	0/1	1/1	0/1	1/1	61

The shortcut experiments reveal a different pattern. At $\sigma = 0.25$, the artificial sensor receives 27–45% of HDF attribution. The relevant-share test detects the resulting loss of alignment in all nine runs up to 100,000 observations, but not at 300,000. At the largest size, the observed relevant score is 0.472, exceeding the null's 95th percentile of 0.417. As the null distribution becomes narrower, a model can remain significantly better than shuffled-label models while relying heavily on a misleading feature.

The shortcut score detects the stronger shortcut in all ten evaluated runs, including at 300,000 observations. At that size, its observed value is 0.249, compared with the null's 95th percentile of 0.211. Under the combined decision rule, HDF is therefore flagged rather than aligned in that run; across all experiments, this is the only case in which the shortcut test changes a verdict that R_m and D_m alone would have given. The weaker shortcut at $\sigma = 0.35$, which receives 6–20% of attribution, is not detected reliably.

The shortcut score also flags TWF in 7 of 30 runs from $3 \cdot 10^4$ observations onward, three of them without an injected sensor, always pointing to air or process temperature with small excess shares (0.03–0.05). The generator draws temperatures independently of tool life, so these flags cannot reflect a genuine shortcut. They likely arise because attribution on TWF observations, which the model captures poorly, is spread more evenly across features than on other failures, and the narrowing null at large n makes this small excess significant. A significant C_m should therefore be read together with the size of the excess share and the identity of the flagged feature.

a) Computational cost: The audit requires $5(K + 1)$ model fits because each real and randomized model is evaluated through five-fold cross-fitting. A single cross-fitted run takes approximately 1.3 seconds for 10^4 observations, 31 seconds for $3 \cdot 10^5$, and 115 seconds for 10^6 observations

on one core, growing roughly linearly with n. With eight cores, the complete $K = 200$ audit takes about three minutes at 10^4 observations and approximately one hour at $3 \cdot 10^5$. The independent randomized runs can be distributed across multiple processors or machines.

F. Comparison with Simpler Audit Designs

We compare the proposed audit with four simpler approaches: global SHAP ranking, a same-model random-observation null, in-sample explanations, and the relevant-share statistic alone. We assess whether each design rejects a shuffled-label model, avoids an overly favorable TWF verdict, and detects a strong shortcut at 300,000 observations.

Table VIII shows that each simpler design misses at least one of these checks. Global feature ranking does not provide a mechanism-level statistical verdict. The naive null approves a shuffled-label model, while in-sample explanations give an optimistic TWF result. The relevant-share test alone can miss strong shortcuts on larger datasets. Together, the results show the value of combining out-of-sample explanations, mechanism-specific attribution measures, and a retrained null. However, the full audit is not guaranteed to detect every type or strength of misleading attribution.

VI. Conclusion

This study evaluated whether SHAP explanations of predictive maintenance models align with known physical failure mechanisms. Using AI4I 2020, we proposed a statistical audit that compares feature attributions with expected failure conditions and shuffled-label models. Despite achieving a test F1 of 0.89, the XGBoost model showed consistent alignment for three failure mechanisms, while tool wear results depended on the reference distribution used by the explanation method, and a neural network showed weaker, explainer-dependent alignment. The audit also revealed that accurate predictions can rely on misleading features and that simpler evaluation methods may fail to detect them. These findings highlight the need to evaluate explanations beyond predictive accuracy to better assess their agreement with known failure mechanisms.

VII. Limitations and Future Work

a) Simulated data: The audit is evaluated on the simulated AI4I 2020 data and on data generated from the same documented process, so it establishes agreement with known simulator rules, not verified physical causes in real systems. We could not address this within the present study, because real maintenance datasets rarely record which mechanism caused each failure. The audit also requires predefined trigger features, which domain experts must supply for real equipment.

b) Reference distribution and explainer: We traced the TWF disagreement between SHAP formulations mainly to the reference distribution (Section V-D), but this does not resolve which reference is correct: verdicts can change with the background, and an oversampled background still leaves a small gap to path-dependent TreeSHAP. Results on the MLP depend on the explainer, and we could not separate the MLP's weaker fit from KernelSHAP's sampling error. LIME was not evaluated, because its surrogate coefficients are not an additive decomposition of the prediction, which the attribution shares assume.

c) Shortcut detection: The shortcut statistic first flags the planted sensor at 35% of the HDF attribution, whereas the relevant-share test already fails at 21%, and no test detects a sensor taking 13% or less. A studentized variant, which scales each feature's excess by its own null spread, was less sensitive still (first flag at 42% rather than 35% shortcut attribution), because the planted sensor already takes a large and variable excess share under the null (mean 0.087, s.d. 0.060). At large n, the statistic also flags small excesses (0.03–0.05) on poorly learned mechanisms, so flags must be read with their size and feature.

d) Statistical scope: Re-tuning the null models widens the null and leaves OSF's relevant share borderline ($p = 0.0498$); re-tuning costs about 50 minutes for $K = 200$, so the primary audit uses carried-over hyperparameters. The bootstrap intervals hold the models fixed. TWF, with 46 observations and a random failure point, and RNF remain hard to assess.

e) Future work: We plan to apply the audit to real data with recorded failure causes, including overlapping faults in the Tennessee Eastman Process [40], [41]; to develop shortcut statistics that are more sensitive without more false flags; and to add complementary statistics that use the whole attribution distribution, such as rank-based or earth mover's distances to the expected feature set, together with dependence-based global sensitivity indices [30] as an attribution-free check for correlated features.

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TABLE VIII
COMPARISON OF AUDIT DESIGNS ON THREE CHECKS. EACH SIMPLER APPROACH OMITS AT LEAST ONE COMPONENT OF THE PROPOSED AUDIT.

Design	Rejects shuffled labels?	Avoids optimistic TWF verdict?	Detects large-data shortcut?
Global SHAP ranking [16]	No statistical verdict	No statistical verdict	No statistical verdict
Naive null	No: all pass ($p = 0.002$)	No: TWF passes	No: HDF passes
In-sample SHAP with shuffled null	Yes	No: TWF passes ($p = 0.004$)	No: $p_C = 0.06$
Relevant share R_m only	Yes	Yes: $p = 0.201$	No: 0 of 1 runs
Full audit	**Yes**	**Yes**	**Yes: 1 of 1 runs**

Note: A successful audit should reject alignment for a model trained on shuffled labels, avoid declaring TWF aligned based only on in-sample explanations, and detect the injected shortcut in the 300,000-observation experiment. Reported p-values refer to the corresponding statistical tests; "0 of 1" and "1 of 1" denote detection outcomes in a single run.

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